

Skill Arena - Phase 1 Part 4

House Challenge Engine & Procedural Generation Specification

Chapter 7 - House Challenge Overview

House Challenges are player-versus-platform competitions. Unlike PvP, House Challenges generate unique content for each player. No two players should receive the same challenge configuration.

House Challenge Objectives

The House Challenge system must: - Remain Fair - Remain Verifiable - Remain Profitable - Prevent Exploitation - Scale Infinitely Every challenge must be unique and auditable.

Challenge Unlock Requirements

Access is controlled through: - XP Level - Skill Rating - Match History - House Reputation House tiers unlock progressively.

House Tiers

Bronze House Silver House Gold House Diamond House Future tiers may be added dynamically. Higher tiers: - Cost More - Reward More - Require Greater Skill

Unique Challenge Generation

House challenges use: Player ID Timestamp Difficulty Tier Secret Server Salt These values generate a unique challenge seed. Challenge generation occurs server-side only.

Procedural Generation Engine

The procedural engine controls: - Maze Size - Path Length - Dead Ends - Branch Count - Objectives - Obstacles - Visibility Modifiers - Timers Generation is infinite.

Similarity Detection Engine

Every generated challenge receives a fingerprint. The system compares: - Solution Path - Structure - Complexity - Difficulty Profile Challenges that are too similar are rejected and regenerated.

Challenge Fingerprinting

Each challenge stores: - Challenge ID - Generation Hash - Difficulty Score - Fingerprint Hash Used to prevent duplicate content.

Difficulty Engine

Difficulty is calculated using: - Path Length - Decision Points - Dead Ends - Obstacles - Objectives - Visibility Restrictions - Lives Available Difficulty Score generated before challenge starts.

Dynamic Timer Engine

Timer is calculated using: - Difficulty Score - Historical Completion Data - House Tier Timer should create a challenging but achievable experience.

House Reputation System

Players earn House Reputation through: - Successful Challenges - Higher Difficulty Wins - Consistency House Reputation has no cap.

Challenge Validation

Before launch: 1. Generate Challenge 2. Calculate Difficulty 3. Check Similarity 4. Create Replay Record 5. Store Audit Record 6. Launch Challenge All validation occurs server-side.

Replay Integration

Every House Challenge stores: - Challenge Seed - Moves - Decisions - Completion Time - Outcome Players may review completed challenges.

Treasury Protection

The House Engine must never create liabilities exceeding available reserves. Maximum payout limits are enforced automatically. Treasury rules always override challenge creation.

House Risk Engine

System monitors: - Success Rate - Failure Rate - Payout Rate - Profitability Challenge frequency and availability may be adjusted to maintain sustainability.

Anti-Exploitation Rules

Players may not: - Replay the same challenge - Reuse challenge seeds - Access generation logic - Predict future challenges All generation systems remain server-side.

Server Authority

The server controls: - Challenge Creation - Difficulty Calculation - Similarity Analysis - Timer Generation - Validation - Rewards - Reputation Client receives display data only.